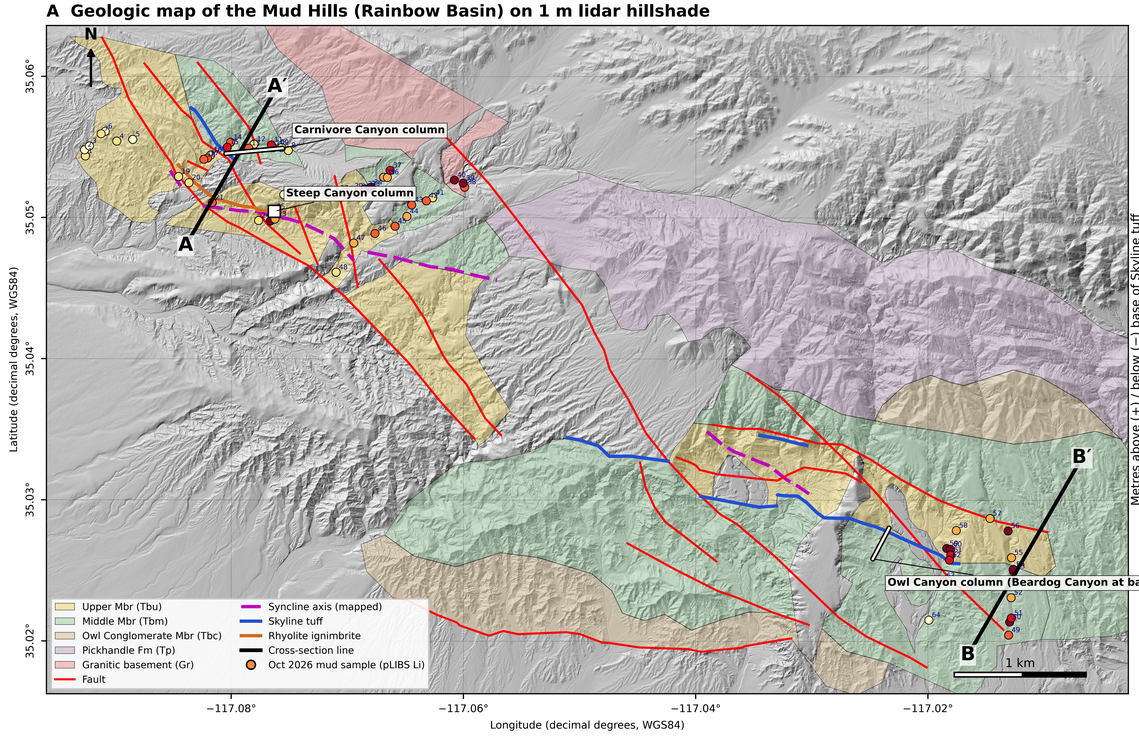
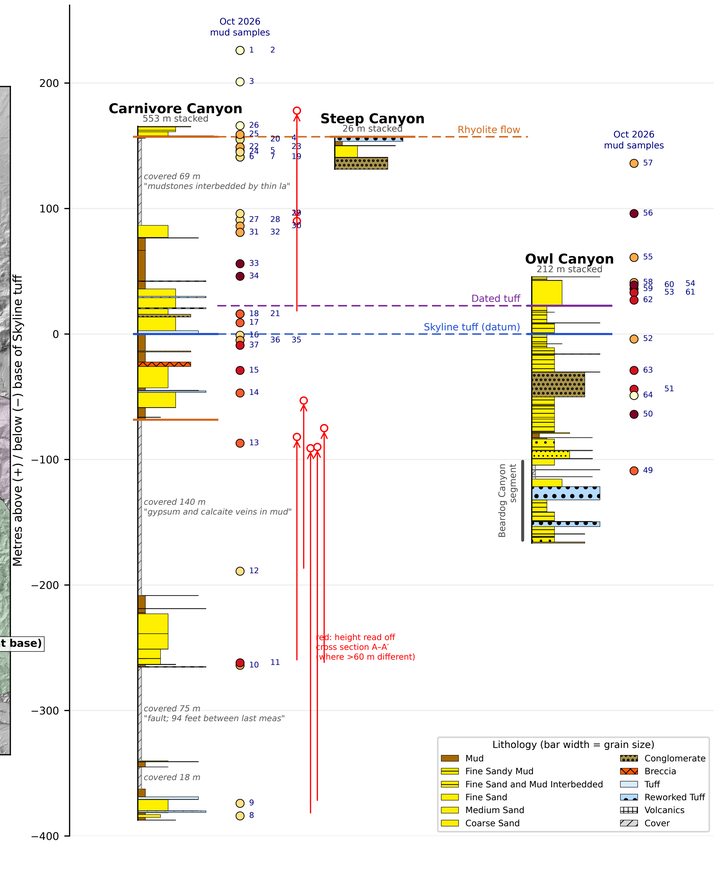
Stratigraphy and structure of the Barstow Formation, Mud Hills, with October 2026 lithium-bearing mudstone samples

GE121B/C · N. Anderson · Field data Mar–Apr 2025 and 23–25 Oct 2026 · pLIBS Li (ppm) colour scale shared by all panels

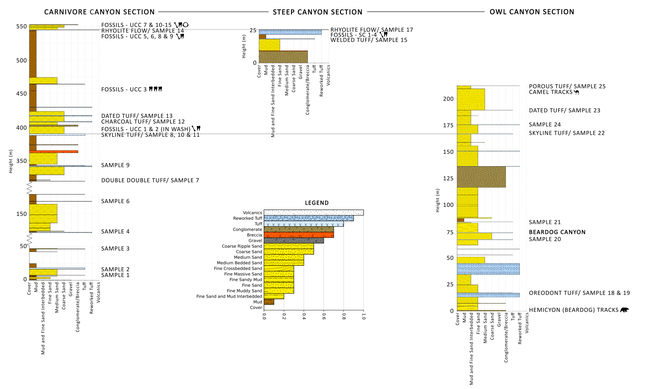
A Geologic map of the Mud Hills (Rainbow Basin) on 1 m lidar hillshade



B Reference columns redrawn on the Skyline-tuff datum



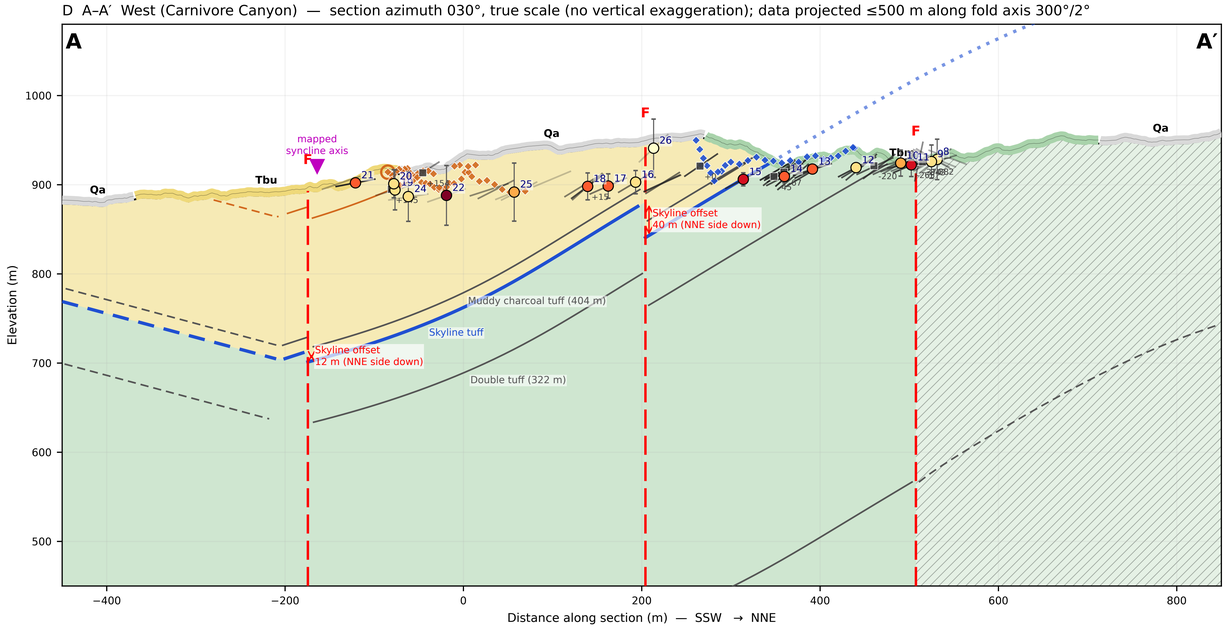
C Fence diagram of the Mud Hills reference sections (Ge121, 2025; ori



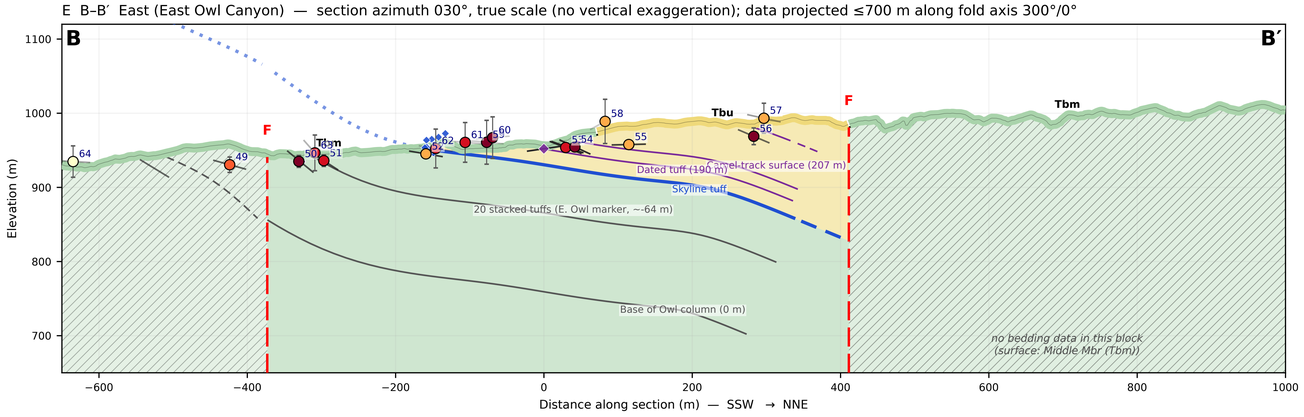
KEY POINTS

- Hung on the Skyline tuff, the columns match the 2025 tie heights: Skyline 387 m (Carnivore) and 167 m (Owl); Dated tuff +23 m; the Carnivore rhyolite flow (Sample 14) sits +157 m above the Skyline and caps the Steep Canyon column.
- 215 m of the 387 m below the Skyline in the Carnivore column is covered interval (beds 17 and 33; bed 17 is logged as a fault). On cross section A–A′ the column base lies only ~80–90 m below the Skyline, so MUD-8 to 12 sit 140–290 m higher than the column heights say (red arrows in panel B). Faulted repetition and the covered intervals are the prime suspects.
- East end: the East Owl and Owl Camp samples fall within ~5–25 m of their assigned heights on B–B′ (median -14 m), and the East Owl marker tuffs fit the Dated-tuff correlation within 15 m (a Skyline correlation fits about as well).
- Every bed measured north of the bounding faults dips toward the trough: the syncline hinge lies at or under the southern fault in the west (A–A′) and under the E–W fault in the east (B–B′); the north limb is unmeasured in the east.
- MUD-21 (assigned +16 m, low confidence) plots near the rhyolite level (~+178 m) on A–A′ and should be re-checked.

D A–A′ West (Carnivore Canyon) — section azimuth 030°, true scale (no vertical exaggeration); data projected ≤500 m along fold axis 300°/2°



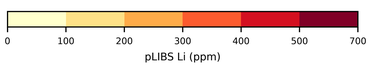
E B–B′ East (East Owl Canyon) — section azimuth 030°, true scale (no vertical exaggeration); data projected ≤700 m along fold axis 300°/0°



Upper Member (above Skyline tuff)
Middle Member (below Skyline tuff)
Block not tied to a marker / no bedding (drawn continuous, offset unknown)
Skyline tuff (constructed)
Other marker horizons, offset by column thickness
Inferred (dip interpolated / assumed, or untied block)

Projected above present topography (eroded)
Mapped fault (drawn vertical; dip & offset not measured)
Bedding, apparent dip (fainter = projected farther)
Skyline tie: map trace / measured outcrop (projected)
Ignimbrite map trace (projected)

Carnivore column tie points (label = m rel. Skyline from column)
East Owl marker tuffs (Dated, camel tracks, 20 stacked tuffs)
Oct 2026 mud sample (colour = pLIBS Li; bar = ± plunge uncertainty)
Mapped syncline axis at surface
Skyline / ignimbrite map trace crosses section line



Sections: true scale, data projected along the fold axis (300°); faults drawn vertical (dips not measured). Columns: bed thicknesses stacked in file order as in pyStrat; covered intervals hatched grey.
See cross_sections/README.md for the construction.